The direction of self-evaluation bias in active learning depends on the metric: accuracy underestimates, macro-F1 overestimates

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Abstract

Practitioners of pool-based active learning routinely evaluate their model on a held-out split of the labels they have collected — because that is all they have. It is folklore that this self-evaluation is biased, since actively sampled data are not i.i.d.; but the direction and size of the bias are rarely measured. We measure both, at scale, with a perfect-oracle protocol that separates a labelable *pool* (50,000 short product descriptions, 621 classes, severe long tail) from a *reserved population* (177,000+ instances) never touched by selection. At each step the collected labels are split 80/20; the same model is scored on the internal test split and on the population. Across five selection policies and two classifiers we find the bias is large, systematic and *metric-dependent*: under uncertainty sampling, internal accuracy *underestimates* deployment accuracy by up to 14 points (the internal test inherits the hard cases), while internal macro-F1 *overestimates* it by up to 34 points (the active sample over-represents rare classes). A random-sampling control shows near-zero gap, validating the instrument. Two corollaries follow. First, *more labels can hurt*: for a logistic-loss classifier, training on the ≈15k actively selected labels beats training on the full 50k pool by 15 macro-F1 points, because the active sample acts as a balanced curriculum — while a per-class-normalized prototype classifier is immune. Second, an ablation of a semantic-diversity selection policy shows its entire continuous-regime gain comes from stratifying by predicted class, yielding a trivial, strong, model-agnostic baseline. Self-evaluated metrics from active learning runs should not be used for deployment decisions in either direction; we quantify why, and what to do instead.

Keywords: active learning; sampling bias; evaluation; class imbalance; macro-F1; text classification.

1 Introduction

Pool-based active learning (AL) buys labels selectively (Settles, 2012; Lewis and Gale, 1994). Its success metric — performance per label — is usually reported on a benchmark with abundant held-out gold data. Deployed AL has no such luxury: the only labeled data available for evaluation are the labels the loop itself collected, and the standard practice is to hold out a fraction of them. The resulting sample is not i.i.d. by construction: uncertainty sampling over-represents boundary cases and rare classes (Attenberg and Provost, 2010; Santos, 2016). That this biases evaluation is known in principle; *how much*, *in which direction*, and *how the direction interacts with the metric* is rarely quantified — yet these are exactly the questions a practitioner needs answered before trusting an internally computed number for a release decision.

We answer them with a deliberately simple protocol (Section 3): a perfect oracle (gold labels), a fixed labelable pool, and a large reserved population that selection can never touch. The gap between internal and population scores, tracked along the full labeling trajectory, *is* the self-evaluation bias — and a random-sampling control arm certifies that the gap is caused by selection, not by the instrument.

Contributions: (i) the first (to our knowledge) large-scale, controlled quantification of AL self-evaluation bias with 621 classes, showing it is **bidirectional and metric-dependent** — internal accuracy underestimates deployment performance while internal macro-F1 overestimates it; (ii) a mechanistic account and empirical demonstration of the *less-is-more* effect — full-pool labeling can be worse than stopping at 30% for loss functions dominated by the natural class distribution — including a classifier (per-class prototype) immune by construction; (iii) an ablation showing a semantic-diversity continuous policy owes its gain entirely to prediction-stratification, exposing a trivial strong baseline; (iv) operational guidance: what internal metrics can and cannot support, and a reserved-set release criterion.

2 Related work

Sampling bias in AL is acknowledged since the field's foundations (Settles, 2012); failure modes of uncertainty sampling under imbalance are documented (Attenberg and Provost, 2010), and bias-aware AL has been studied theoretically (Santos, 2016). Density and cluster-based policies (Settles and Craven, 2008; Nguyen and Smeulders, 2004; Dasgupta and Hsu, 2008) motivate the diversity component we ablate. Metric behavior under imbalance (micro vs. macro aggregation) is classical (Sokolova and Lapalme, 2009). What is missing in this literature — and what we contribute — is a controlled, trajectory-level measurement of the *evaluation* bias (not the training bias) with its metric dependence, at a class cardinality where the two metrics dissociate dramatically. The renewed practical interest comes from LLM-oracle pipelines (Gilardi et al., 2023; Bayer and Reuter, 2024), where cheap labels make full-trajectory experiments affordable and self-evaluation is the default telemetry.

3 Protocol: internal vs. population evaluation

Let D be a deduplicated corpus, shuffled once with a fixed seed and split into a **pool** P ($|P| = 50{,}000$), from which the learner may request labels, and a **population** Q ($|Q| = 181{,}490$), reserved in full. The oracle is perfect (gold labels), removing annotation noise as a confounder. The loop starts from a random batch and iterates: train on the collected set L_t, select the next batch of 500 from $P \setminus L_t$, until the pool is exhausted. At every step, L_t is split 80/20 (stratified when feasible, fixed seed); the model is trained on the 80% and scored twice with identical weights: on the **internal test** (the 20% — what a practitioner without reserved data can measure) and on the **population** Q (deployment performance). The signed gap $\Delta_m(t) = m_{\text{int}}(t) - m_{\text{ext}}(t)$ for metric m is the self-evaluation bias.

Control arm. Under random selection, L_t is an i.i.d. sample of P (and P of D), so Δ should vanish up to noise. A non-zero Δ under random selection would indicate an instrument artifact (split leakage, size effects); observing $\Delta \approx 0$ validates the protocol, and every deviation under a non-random policy is then attributable to selection.

4 Experimental setup

Data and task. 231,490 deduplicated short product descriptions from Brazilian fiscal receipts (4–50 characters, uppercase, abbreviations), 649 classes with at least 2 instances, severe long tail. **Classifiers**, chosen as extremes of loss geometry: *SGD* — logistic regression on binary TF–IDF via stochastic gradient descent, loss dominated by the empirical class distribution; *PVBin* — a per-class prototype model over binary term vectors (Darú, 2024), normalized per class by construction. **Policies**: uncertainty (entropy); random (control); a semantic-diversity policy (density-representative selection over SBERT embeddings with lexical-novelty tie-breaking, DRI-SL); the same policy with groups given by the classifier's *predicted classes* (DRI-SL-C); and the ablation of the latter without lexical novelty — equivalent to *stratified sampling by predicted class*. Batch 500, budget to pool exhaustion, seed fixed. [TODO: campanha de 8 sementes em fila — converter achados em inferenciais (IC entre sementes + Wilcoxon pareado)]

5 Results

5.1 Selector comparison on the population

Table 1: Population macro-F1 by policy and classifier (single seed, descriptive). Saturation: smallest $|L|$ reaching 95% of the arm's own ceiling.

Clf.	Policy	Ceiling	Saturation	F1@10k	F1@20k
SGD	Entropy	**0.591**	**8,000**	0.565	0.574
SGD	Stratif. by prediction	0.555	10,000	0.533	0.543
SGD	DRI-SL-C	0.491	15,500	0.412	0.486
SGD	Random	0.459	16,500	0.391	0.449
SGD	DRI-SL	0.441	41,500	0.310	0.365
PVBin	Entropy	0.529	19,000	0.438	0.509
PVBin	Stratif. by prediction	0.528	18,000	0.444	0.498
PVBin	DRI-SL-C	0.525	39,500	0.349	0.453
PVBin	Random	0.530	40,000	0.348	0.423
PVBin	DRI-SL	0.527	45,500	0.284	0.356

Entropy dominates at scale (Table 1): it saturates at 95% of its ceiling with 8,000 labels for SGD (19,000 for PVBin), half to a quarter of the random requirement. The semantic-diversity policy (DRI-SL), designed for the zero-label cold start, is *worse than random* as a continuous policy; guiding its groups by predicted classes (DRI-SL-C) repairs it (ceiling 0.441 → 0.491 for SGD, saturation 2.7× earlier).

5.2 The ablation: stratification is the active ingredient

Removing the lexical-novelty term from DRI-SL-C — leaving plain stratified sampling by predicted class — *improves* it on both classifiers (SGD: ceiling 0.555 vs. 0.491, saturation 10k vs. 15.5k; PVBin: saturation 18k vs. 39.5k, tying entropy). The diversity heuristic, valuable at cold start where no model exists, is counterproductive in the continuous regime: it privileges lexically atypical exemplars when the macro metric pays for balanced class coverage. The practical corollary is a baseline: *predict, stratify, sample* — model-agnostic, no embeddings, no tuning — that captures most of the uncertainty-sampling gain and nearly all of the diversity-policy gain.

5.3 The bias is bidirectional and metric-dependent

The control behaves: under random selection the internal–external accuracy gap is $\Delta \approx 0$ along the whole trajectory. Under entropy selection, internal **accuracy underestimates** the population value by up to **14 points**: the internal test inherits the boundary-heavy, hard-case distribution that uncertainty sampling selects. Internal **macro-F1 overestimates** the population value for *every* non-random policy — by up to +34 **points** early in the DRI-SL trajectory: the active sample gives rare classes internal test support far above their population share, so per-class scores are computed on flattering supports. Neither direction is usable as a deployment estimate, and the two errors do not cancel: a practitioner reading internal accuracy will under-deploy (the model is better than it looks); one reading internal macro-F1 will over-promise tail quality (it is worse than it looks).

5.4 More labels can hurt: the active sample as curriculum

For SGD under entropy selection, population macro-F1 peaks at 0.59 around 15k labels and *falls* to 0.44 when the entire pool is labeled: the final 35k labels, drawn increasingly from the natural (imbalanced) distribution, drag the logistic loss toward the head classes. The actively selected subset acts as a balanced curriculum — and labeling everything undoes it. PVBin, whose per-class prototypes normalize away the class distribution, is immune (ceiling ≈ flat across policies). Beyond waste, additional labels can be *harmful* to macro metrics for distribution-sensitive losses; remedies are early stopping by stagnation, reweighting, or per-class-normalized models.

6 Operational guidance

(i) Maintain a reserved evaluation set from day one — the bias cannot be corrected post hoc from internal data alone, since its size and sign depend on the (policy, metric, model) triple. (ii) Use internal metrics only for *relative* decisions inside one run (stagnation-based stopping), never for absolute release decisions. (iii) If a reserved set is impossible, prefer internal accuracy over internal macro-F1 for go/no-go: it errs conservative. (iv) For macro objectives with distribution-sensitive losses, treat the labeled-set composition as part of the model: stopping earlier, subsampling or reweighting may beat labeling more.

7 Limitations

Single dataset and domain (short retail text; the mechanism — support inflation of rare classes — is general, but effect sizes are not); single seed per cell in this draft (a multi-seed campaign with between-seed intervals and paired Wilcoxon tests is queued [TODO: atualizar com 8 sementes]); perfect oracle by design (noisy oracles would add a second bias source; our companion work on LLM oracles addresses it separately, keeping this paper's question clean).

8 Conclusion

Self-evaluation on actively collected data is not merely noisy — it is systematically wrong in a direction that depends on the metric: accuracy reads low, macro-F1 reads high, by double-digit margins at 621-class granularity. A random control certifies the effect is caused by selection. The same experiments yield a practical bonus (stratified-by-prediction as a trivial strong policy) and a warning (labeling the full pool can damage macro performance for distribution-sensitive losses). All trajectories, code and per-step artifacts are released.

Artifacts. `github.com/GHDaru/activelearning` (`experiments/e6population`).

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